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Spanning Trees Wait for Triangles

Entanglement, light cones, and the homology of the unresolved bit

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Abstract

Why does quantum entanglement exhibit exact correlations across spacelike separation without violating the relativistic speed of light c? Standard presentations often treat the no-signaling theorem as a fortunate algebraic cancellation: measuring one particle supposedly collapses the distant state instantaneously, yet tracing out the first particle's Hilbert space happens to erase any locally detectable change. In this paper, we replace that picture of instantaneous influence and miraculous cancellation with an elementary homological identity over $\mathbb{F}_2$ lifted to the Pauli stabilizer formalism.

We prove that a bipartite or multipartite entangled state prepared in the shared causal past is a **relational spanning tree**: its incidence operator B fixes all relative parities ($y = Bx$, dimension $\text{rank}(B) = n - c$) while leaving the global component orientation ($\ker B$, dimension c) completely unresolved. From this decomposition, two physical consequences follow immediately. First, **no-signaling is a gauge invariance theorem**: any local observable at a spacelike-separated vertex transforms non-trivially under the unresolved orientation group $\ker B$, forcing its marginal expectation value to vanish identically ($\rho_B = I/2$) regardless of whether a distant observer has anchored a gauge root. Second, **the speed of light c is the propagation rate of cycle closure** ($\ker B^T$): operational verification of a correlation, syndrome extraction, or unlocking a teleported quantum state requires converting an open spanning tree ($m = n - 1$, $\dim\ker B^T = 0$) into a closed homological cycle ($m = n$, $\dim\ker B^T = 1$) at an intersection vertex in the overlapping future light cone $J^+(A) \cap J^+(B)$. Finally, by extending the incidence matrix to the symplectic binary space $\mathbb{F}_2^{2n}$, we show why quantum stabilizer states exceed classical hidden-variable trees: enforcing simultaneous cycle closures across non-commuting bases ($Z \otimes Z$ and $X \otimes X$) produces an obstructed global section in which the relational edges exist without underlying vertex assignments.

THE CENTRAL THESIS

Spacelike entanglement does not transmit a relationship across space; it preserves a settled edge ($y_{AB} = x_A \oplus x_B$) from the shared past while leaving the global vertex orientation ($\ker B$, 1 bit) unresolved. Faster-than-light signaling fails because half of an XOR equation is exact white noise ($\rho_B = I/2$). The speed of light c does not bound the existence of a correlation; it bounds the rate at which an open spanning tree can be closed into a verifiable triangle ($\ker B^T$).

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1 Introduction: The False Paradox of Instantaneous Collapse

For ninety years, discussions of the Einstein–Podolsky–Rosen (EPR) paradox [1] and Bell's theorem [2] have oscillated between two uncomfortable narratives. In the first narrative, a local measurement by Alice at vertex A exerts an instantaneous, superluminal effect on Bob's subsystem at spacelike-separated vertex B, collapsing a shared wavefunction across arbitrary distances; relativistic causality is then rescued at the last moment by the no-signaling theorem, which shows that Bob's reduced density matrix $\rho_B = \mathrm{Tr}_A(\rho_{AB})$ remains invariant under Alice's trace-preserving local operations. While mathematically unassailable, this account leaves an explanatory gap: why should nature permit an instantaneous physical state change across space while conspiring to hide it behind a partial trace?

In the second narrative, one attempts to explain the correlations by positing that the individual outcomes (x_A, x_B) were already locally selected at the source S in the shared past. Bell's inequality [2], the CHSH game [3], and the Greenberger–Horne–Zeilinger (GHZ) parity contradiction [4, 5] decisively refute that classical completion: no joint probability distribution over pre-assigned local vertex values can reproduce the correlations of entangled quantum states across non-commuting measurement bases.

In a companion note [9], we explored a third structural possibility: *a system can possess an exact, completely settled relational specification while leaving an explicit symmetry degree of freedom unresolved*. When an observer measures all pairwise XOR relations $y_{uv} = x_u \oplus x_v$ along a connected graph $G = (V, E)$ of n binary variables, the edge-vertex incidence matrix $B \in \mathbb{F}_2^{m \times n}$ has rank $n - 1$ and kernel dimension $\dim\ker B = 1$. Every relative agreement or disagreement in the network is fixed, yet exactly one global orientation bit ($x \mapsto x \oplus \mathbf{1}$) remains undetermined. Furthermore, when the graph contains cycles, the left nullspace $\ker B^T$ of dimension $m - n + c$ allows relations to verify one another ($y_{AB} \oplus y_{BC} \oplus y_{AC} = 0$) without unmasking a single vertex identity.

In this paper, we show that this exact homological split between the **cut space** (im B, dimension $n - c$), the **unresolved orientation kernel** ($\ker B$, dimension c), and the **cycle verification space** ($\ker B^T$, dimension $m - n + c$) provides a transparent, non-paradoxical foundation for relativistic quantum entanglement:

1. **Spacelike Entanglement is an Open Spanning Tree (**$\dim\ker B = 1, \dim\ker B^T = 0$**):** When a source S prepares an entangled pair and sends the subsystems to spacelike-separated regions A and B, the settled stabilizer generators fix the relative edge parity y_{AB} in the causal past. No individual vertex states x_A, x_B are selected.

2. **No-Signaling is the** $\ker B$ **Gauge Screen:** When Alice measures her subsystem at A, she does not broadcast a physical perturbation across space; she anchors a local gauge root $x_A \in \mathbb{F}_2$ for her record. Because Bob at B lacks access to x_A, his local observables remain averaged over the $\ker B$ symmetry orbit ('01' $\leftrightarrow$ '10'), yielding $\rho_B = I/2$ identically.

3. **Light Speed (**c**) Bounds Homological Cycle Closure (**$\dim\ker B^T : 0 \to 1$**):** An open spanning tree cannot verify its own parity or extract a relative phase without a cycle-closing edge. Converting the unresolved global bit into an operational verification (such as auditing a Bell violation or decoding a teleported state) requires classical syndrome records from A and B to propagate along causal worldlines at speed $\leq c$ to a common vertex M in the overlapping future light cone $J^+(A) \cap J^+(B)$, closing the homological triangle.

2 Homology of a Spacetime Diamond Over $\mathbb{F}_2$

2.1 The Three Fundamental Subspaces of Relational Observation

Let $G = (V, E)$ be a finite undirected graph with $n = |V|$ vertices, $m = |E|$ edges, and c connected components. Assign to each vertex $v \in V$ a binary state variable $x_v \in \mathbb{F}_2 = \{0, 1\}$, forming the vertex state vector $x \in \mathbb{F}_2^n$. Assign to each edge $e = (u, v) \in E$ the relational parity observable

$$y_e = x_u \oplus x_v \pmod 2, \tag{1}$$

which equals 0 when the endpoints agree and 1 when they disagree. In the language of algebraic topology over $\mathbb{F}_2$, vertex assignments $x \in C^0(G; \mathbb{F}_2) \cong \mathbb{F}_2^n$ are 0-cochains, edge records $y \in C^1(G; \mathbb{F}_2) \cong \mathbb{F}_2^m$ are 1-cochains, and the $m \times n$ edge-vertex incidence matrix B (where row $e = (u, v)$ has 1 in columns u and v, and 0 elsewhere) is the coboundary operator $\delta_0 : C^0 \to C^1$ mapping vertex states to edge relations:

$$y = Bx \pmod 2. \tag{2}$$

Theorem 2.1 (Homological Decomposition of Relational Observation)**.** *For any graph $G = (V, E)$ with n vertices, m edges, and c connected components, the incidence operator $B \in \mathbb{F}_2^{m \times n}$ induces an exact decomposition of the vertex state space $\mathbb{F}_2^n$ and the edge measurement space $\mathbb{F}_2^m$:*

$$\textit{Vertex Space } (n \textit{ bits}): \quad n = \underbrace{(n-c)}_{\textit{Resolved Skeleton } (\textit{rank } B)} + \underbrace{c}_{\textit{Unresolved Orientation } (\dim\ker B)}, \tag{3}$$

$$\textit{Edge Space } (m \textit{ bits}): \quad m = \underbrace{(n-c)}_{\textit{Spanning Forest } (\textit{dim im } B)} + \underbrace{(m-n+c)}_{\textit{Cycle Checks } (\dim\ker B^T)}. \tag{4}$$

Specifically:

1. ***Unresolved Gauge Kernel (***$\ker B = H^0(G; \mathbb{F}_2)$***):*** *The nullspace* $\ker B = \{z \in \mathbb{F}_2^n : Bz = 0\}$ *is spanned by the c indicator vectors $\mathbf{1}_{V_1}, \ldots, \mathbf{1}_{V_c}$ of the connected components of G. For any realizable edge record $y \in \text{im } B$, the solution set $\{x \in \mathbb{F}_2^n : Bx = y\}$ is an affine coset $x_0 + \ker B$ of cardinality 2^c.*

2. ***Cycle Verification Space (***$\ker B^T = H_1(G; \mathbb{F}_2)$***):*** *An edge vector $y \in \mathbb{F}_2^m$ belongs to im B if and only if $Cy = 0 \pmod 2$, where the rows of $C \in \mathbb{F}_2^{(m-n+c)\times m}$ form a basis for the cycle space* $\ker B^T$*. Every simple cycle $\gamma \subseteq E$ enforces the vertex-free parity check $\bigoplus_{e\in\gamma} y_e = 0$.*

Proof. For a vector $z \in \mathbb{F}_2^n$, the condition $Bz = 0$ holds if and only if $z_u \oplus z_v = 0$ (i.e., $z_u = z_v$) for every edge $(u, v) \in E$. By induction along paths, z is constant on each connected component V_i ($i = 1, \ldots, c$). Since the component indicator vectors $\mathbf{1}_{V_1}, \ldots, \mathbf{1}_{V_c}$ are disjointly supported, they are linearly independent over $\mathbb{F}_2$, giving $\dim\ker B = c$. By the rank-nullity theorem, $\text{rank}(B) = \dim\text{im } B = n - c$, establishing (3). Applying rank-nullity to the transpose $B^T \in \mathbb{F}_2^{n\times m}$ yields $\dim\ker B^T = m - \text{rank}(B^T) = m - (n-c) = m-n+c$, establishing (4). Around any cycle $\gamma = ((v_1, v_2), (v_2, v_3), \ldots, (v_k, v_1))$, the indicator vector $w_\gamma \in \mathbb{F}_2^m$ sums two 1s in the column of each vertex $v_j \in \gamma$, so $B^T w_\gamma = 0 \pmod 2$. Since there are $m - n + c$ fundamental cycles with respect to any spanning forest $F \subseteq E$ (one for each chord $e \in E \setminus F$), these fundamental cycles form a basis for $\ker B^T$. □

2.2 Spacelike Separation as an Open Tree vs. Light-Cone Merge as a Closed Cycle

Now embed Theorem 2.1 into relativistic spacetime. Consider a standard bipartite correlation experiment with three spacetime regions:

- **Preparation Event S (Causal Past):** Located in the shared past light cone $J^-(A) \cap J^-(B)$. At S, an interaction prepares two subsystems and fixes their relative parity $y_{AB} = x_A \oplus x_B \in \mathbb{F}_2$ (equivalent to enforcing the path relation $y_{SA} \oplus y_{SB} = y_{AB}$ via the source), while leaving the root orientation $x_S \in \mathbb{F}_2$ uniform and unmeasured.
- **Spacelike Measurement Events A and B:** The two subsystems propagate along causal worldlines $S \to A$ and $S \to B$ at speed $\leq c$ to spacelike-separated regions A and B ($A \notin J^{\pm}(B)$).
- **Verification / Decoding Event M (Causal Future):** Located in the overlapping future light cone $J^+(A) \cap J^+(B)$, where classical records from A and B can be brought together by light-speed signals $A \to M$ and $B \to M$.

Spacetime Slice	Graph Topology	Active n	Edges
Stage I: Spacelike Slice Σ_{space} ($A \perp B$)	Open Path / Tree ($A \xleftarrow{y_{AB}} B$)	2	1
Stage II: Future Light-Cone Merge $M \in J^+(A) \cap J^+(B)$	Closed Cycle / Triangle ($S - A - B$)	3	3

Table 1: Topological transition of a bipartite correlation across a relativistic spacetime diamond. On any spacelike hypersurface separating A and B, the relational graph is a tree ($m = n - 1$), leaving one unresolved orientation bit and zero cycle checks. Closing the loop requires causal edges to meet in $J^+(A) \cap J^+(B)$ at speed $\leq c$.

As summarized in Table 1, on any spacelike hypersurface Σ_{space} intersecting A and B, the effective graph on the active endpoints $\{A, B\}$ has $n = 2$ vertices, $m = 1$ relational edge (y_{AB} established via S), and $c = 1$ connected component. By Theorem 2.1:

$$\operatorname{rank}(B) = 2 - 1 = 1, \qquad \dim\ker B = 1, \qquad \dim\ker B^T = 1 - 2 + 1 = 0. \tag{5}$$

Equation (5) captures the entire causal structure of spacelike entanglement in three numbers:

1. $\operatorname{rank}(B) = 1$: The relationship between A and B is **completely settled** (1 bit of relational structure).
2. $\dim\ker B = 1$: Exactly **one global orientation bit remains unresolved**. The compatible state space at (A, B) is the two-element coset $\Omega(y_{AB}) = \{(0, y_{AB}), (1, 1 \oplus y_{AB})\}$.
3. $\dim\ker B^T = 0$: There are **zero closed cycles** on the spacelike slice. Neither A nor B can perform a consistency check or extract the relative parity locally.

Proposition 2.2 (The Classical ker B No-Signaling Screen)**.** *Let $(X_A, X_B) \in \mathbb{F}_2^2$ be uniformly distributed over the relational coset $\Omega(y_{AB}) = \{x \in \mathbb{F}_2^2 : x_A \oplus x_B = y_{AB}\}$ for a fixed edge parity $y_{AB} \in \mathbb{F}_2$. Then:*

1. *The joint entropy is $H(X_A, X_B \mid Y_{AB} = y_{AB}) = \dim\ker B = 1$ bit, while the conditional entropy of either vertex given the other is $H(X_B \mid X_A, Y_{AB}) = 0$ bits.*
2. *The marginal distribution of X_B is independent of the settled edge parity y_{AB} and invariant under the global flip action of $\ker B = \{00, 11\}$:*

$$\mathbb{P}(X_B = 0 \mid Y_{AB} = 0) = \mathbb{P}(X_B = 0 \mid Y_{AB} = 1) = \frac{1}{2}, \qquad I(X_B; Y_{AB}) = 0 \text{ bits}. \tag{6}$$

Consequently, observing $X_A = x_A$ at vertex A selects one of the two coset representatives via $x_B = x_A \oplus y_{AB}$, but until x_A is transported along a causal edge to B, the marginal entropy at B remains $H(X_B \mid Y_{AB}) = 1$ bit.

Proof. For $y_{AB} = 0$, the uniform distribution on $\Omega(0) = \{00, 11\}$ gives $\mathbb{P}(X_B = 0) = \mathbb{P}(\{00\}) = 1/2$ and $\mathbb{P}(X_B = 1) = \mathbb{P}(\{11\}) = 1/2$. For $y_{AB} = 1$, the uniform distribution on $\Omega(1) = \{01, 10\}$ gives $\mathbb{P}(X_B = 0) = \mathbb{P}(\{10\}) = 1/2$ and $\mathbb{P}(X_B = 1) = \mathbb{P}(\{01\}) = 1/2$. Thus $X_B \sim$ Bernoulli$(1/2)$ regardless of y_{AB}, yielding $I(X_B; Y_{AB}) = H(X_B) - H(X_B \mid Y_{AB}) = 1 - 1 = 0$. □

Proposition 2.2 exposes why the phrase "measuring A changes B" is a category error. When Alice observes $X_A = 0$ at A, she does not alter the edge parity y_{AB} (which was fixed at S in the past), nor does she modify the marginal statistics at B. She simply learns which element of the two-element gauge orbit $\Omega(y_{AB}) = \{01, 10\}$ indexes their shared history. For Bob at B, possessing one variable of an unresolved two-variable XOR equation $x_A \oplus x_B = y_{AB}$ with an unknown uniform partner x_A is algebraically indistinguishable from holding a fair coin.

3 Quantum Lift: Stabilizer Parity and the Symplectic Gauge Screen

A reader familiar with Bell's theorem will immediately raise the essential physical question: *In what exact sense does a quantum Bell state differ from the classical parity coset $\Omega(y_{AB})$ of Proposition 2.2?*

The answer is that a classical tree fixes the incidence relation $y = Bx$ in a **single commutative basis** (Z), whereas a quantum stabilizer state fixes the relational coboundary simultaneously across **two non-commuting conjugate bases** (Z and X). Let us formalize this lift and prove that both the ker B no-signaling screen and the ker B^T light-cone cycle condition extend identically to the quantum regime.

3.1 Symplectic Incidence Matrices for Stabilizer States

Recall that the n-qubit Pauli group $\mathcal{P}_n$ (modulo overall phases $\{\pm 1, \pm i\}$) is isomorphic to the $2n$-dimensional binary symplectic vector space $\mathbb{F}_2^{2n}$ [7]. Each Pauli operator $P = i^\delta \bigotimes_{v=1}^n X_v^{a_v} Z_v^{b_v}$ is represented by a binary row vector $(a \mid b) \in \mathbb{F}_2^{2n}$, where $a, b \in \mathbb{F}_2^n$. Two Pauli operators $P_1 \leftrightarrow (a_1 \mid b_1)$ and $P_2 \leftrightarrow (a_2 \mid b_2)$ commute if and only if their symplectic inner product vanishes over $\mathbb{F}_2$:

$$\langle (a_1 \mid b_1), (a_2 \mid b_2) \rangle_{\text{symp}} = a_1 b_2^T \oplus b_1 a_2^T = 0 \pmod 2. \tag{7}$$

Now consider a connected graph $G = (V, E)$ with n vertices, $m = n - 1$ spanning-tree edges, and edge-vertex incidence matrix $B \in \mathbb{F}_2^{(n-1)\times n}$ of rank $n - 1$. We can associate to G two natural families of relational Pauli stabilizers:

1. **Z-Parity Edge Checks ($n-1$ generators):** For each edge $e=(u,v)\in E$, define the weight-2 parity operator

$$S_e^{(Z)} = Z_u \otimes Z_v \quad \longleftrightarrow \quad (0 \mid b_e) \in \mathbb{F}_2^{2n}, \tag{8}$$

where $b_e \in \mathbb{F}_2^n$ is the e-th row of the incidence matrix B. Measuring $S_e^{(Z)}$ with outcome $(-1)^{y_e^{(Z)}}$ resolves the Z-basis XOR relation $x_u^{(Z)} \oplus x_v^{(Z)} = y_e^{(Z)}$ without measuring Z_u or Z_v individually.

2. **Global Conjugate Orientation Stabilizer (1 generator):** The unresolved orientation kernel $\ker B = \mathrm{span}(\mathbf{1}_n)$ of the Z-incidence matrix generates the global bit-flip operator across all n vertices:

$$S_{\text{global}}^{(X)} = \bigotimes_{v=1}^{n} X_v \quad \longleftrightarrow \quad (\mathbf{1}_n \mid 0) \in \mathbb{F}_2^{2n}. \tag{9}$$

Notice how the homology of the graph G governs the commutation of (8) and (9): for every edge $e=(u,v)\in E$, the symplectic inner product between the edge check $S_e^{(Z)}$ and the global flip $S_{\text{global}}^{(X)}$ is

$$\langle (0 \mid b_e), (\mathbf{1}_n \mid 0) \rangle_{\text{symp}} = b_e \mathbf{1}_n^T = 1 \oplus 1 = 0 \pmod 2, \tag{10}$$

precisely because $\mathbf{1}_n \in \ker B$! That is:

> *The global X-flip operator $\bigotimes_{v=1}^n X_v$ commutes with every Z-edge parity check Z_uZ_v for the exact algebraic reason that flipping all vertices simultaneously preserves every edge XOR ($B\mathbf{1}_n = 0$).*

For $n=2$ vertices connected by a single edge $e=(A,B)$ with incidence matrix $B = \begin{pmatrix}1 & 1\end{pmatrix}$, the stabilizer generators $S_e^{(Z)} = (-1)^{y_Z} Z_A Z_B$ and $S_{\text{global}}^{(X)} = (-1)^{y_X} X_A X_B$ specify the four maximally entangled Bell states:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (y_Z=0,\, y_X=0), \qquad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \quad (y_Z=0,\, y_X=1), \tag{11}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad (y_Z=1,\, y_X=0), \qquad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \quad (y_Z=1,\, y_X=1). \tag{12}$$

More generally, for any connected n-vertex tree, the $n-1$ independent edge checks $S_e^{(Z)} = (-1)^{y_e} Z_u Z_v$ together with the single global orientation check $S_{\text{global}}^{(X)} = (-1)^{y_X} \bigotimes_{v=1}^n X_v$ form a maximal abelian stabilizer group $\mathcal{S}$ of n independent generators, uniquely fixing an n-qubit GHZ-class state $|\Psi_G(y_Z, y_X)\rangle$.

3.2 Theorem: No-Signaling as Non-Invariance Under the Orientation Kernel $\ker B$

We now prove that the quantum no-signaling property is not an accidental cancellation of partial traces, but a direct consequence of the fact that every local observable fails to commute with the unresolved orientation symmetry.

Theorem 3.1 (Quantum Gauge-Screen Theorem)**.** *Let $G=(V,E)$ be a connected graph with $n \geq 2$ vertices and incidence matrix $B \in \mathbb{F}_2^{m\times n}$ of rank $n-1$, and let ρ be the density matrix of any quantum state stabilized by the Z-edge parity checks $\{(-1)^{y_e} Z_u Z_v\}_{e\in E}$ and the global X-orientation stabilizer $(-1)^{y_X} \bigotimes_{v\in V} X_v$. Then:*

1. ***Local Maximal Mixedness:*** *For every vertex $w \in V$, every non-identity single-qubit Pauli observable $O_w \in \{X_w, Y_w, Z_w\}$ has expectation value $\mathrm{Tr}(\rho O_w) = 0$. Consequently, every single-vertex reduced density matrix is maximally mixed:*

$$\rho_w = \mathrm{Tr}_{V\setminus\{w\}}(\rho) = \frac{I_2}{2}, \qquad S(\rho_w) = 1 \text{ bit}. \tag{13}$$

2. ***Invariance Under Remote Projective Measurement (No-Signaling):*** *Suppose an observer at a subset of vertices $A \subsetneq V$ performs an arbitrary local projective measurement (or trace-preserving quantum channel $\mathcal{E}_A$) and does not transmit the classical outcome record s_A to vertex $w \in V \setminus A$. Then the post-measurement ensemble density matrix ρ'_w at w remains identically $\frac{I_2}{2}$.*

Proof. To prove Part 1, recall that if $U \in \mathcal{S}$ is any stabilizer of ρ (so $U\rho U^\dagger = \rho$) and O is any operator that anticommutes with U ($UO = -OU$, or $UOU^\dagger = -O$), then by cyclicity of the trace,

$$\mathrm{Tr}(\rho O) = \mathrm{Tr}(U\rho U^\dagger O) = \mathrm{Tr}(\rho U^\dagger O U) = -\mathrm{Tr}(\rho O) \implies \mathrm{Tr}(\rho O) = 0. \tag{14}$$

Now test the three local Pauli operators at vertex $w \in V$:

- For $O_w = Z_w$: Its binary vector $(0 \mid e_w)$ has odd weight $e_w \mathbf{1}_n^T = 1 \neq 0 \pmod 2$, so $Z_w \notin \mathrm{rowspace}(B)$. Equivalently, Z_w anticommutes with the global orientation stabilizer $S^{(X)}_{\text{global}} = (-1)^{y_X} \bigotimes_{v\in V} X_v$ (which flips $Z_w \mapsto -Z_w$). Thus $\mathrm{Tr}(\rho Z_w) = 0$.
- For $O_w = X_w$ or $O_w = Y_w$: Because G is connected and $n \geq 2$, vertex w is incident to at least one edge $e = (w, u) \in E$. The local operator X_w (or Y_w) anticommutes with the edge parity stabilizer $S^{(Z)}_e = (-1)^{y_e} Z_w Z_u$ at endpoint w while acting as the identity at u. Thus $\mathrm{Tr}(\rho X_w) = \mathrm{Tr}(\rho Y_w) = 0$.

Expanding $\rho_w = \frac{1}{2}\big(I_2 + \mathrm{Tr}(\rho X_w)X + \mathrm{Tr}(\rho Y_w)Y + \mathrm{Tr}(\rho Z_w)Z\big)$ yields $\rho_w = I_2/2$.

For Part 2, suppose Alice at region A measures a local Pauli observable M_A (such as Z_u for $u \in A$) with spectral projectors Π_0, Π_1 ($\Pi_0 + \Pi_1 = I_A$). Conditional on outcome $s_A \in \{0, 1\}$, the post-measurement state is $\rho^{(s_A)} = \frac{\Pi_{s_A}\rho\Pi_{s_A}}{\mathrm{Tr}(\rho\Pi_{s_A})}$. In this conditioned branch, the stabilizer $S^{(X)}_{\text{global}}$ (which fails to commute with Z_u) is replaced by $(-1)^{s_A} Z_u$, anchoring the global Z-orientation root at u and fixing $(-1)^{s_A \oplus y_{uw}} Z_w$ along the tree path from u to w. However, for an observer at $w \notin A$ who has not received the classical bit s_A, the accessible state is the average over the unresolved bit $s_A \in \mathbb{F}_2$:

$$\rho'_w = \sum_{s_A\in\{0,1\}} \mathbb{P}(s_A)\mathrm{Tr}_{V\setminus\{w\}}\big(\rho^{(s_A)}\big) = \mathrm{Tr}_{V\setminus\{w\}}\Big(\sum_{s_A\in\{0,1\}} (\Pi_{s_A}\otimes I_B)\rho(\Pi_{s_A}\otimes I_B)\Big) = \mathrm{Tr}_{V\setminus\{w\}}(\rho) = \frac{I_2}{2}, \tag{15}$$

where we used $[\Pi_{s_A} \otimes I_B, I_A \otimes O_w] = 0$ and $\sum_{s_A} \Pi^2_{s_A} = I_A$. □

Look closely at what the proof of Theorem 3.1 reveals about the physics:

1. Why does $\mathrm{Tr}(\rho Z_w) = 0$? Because Z_w asks for the **absolute vertex orientation**, which anticommutes with the unresolved global flip symmetry $S^{(X)}_{\text{global}} \leftrightarrow \ker B$.
2. Why does $\mathrm{Tr}(\rho X_w) = 0$? Because X_w flips a **single vertex without flipping its neighbor**, which violates the settled edge parity $S^{(Z)}_e = Z_w Z_u \leftrightarrow \mathrm{im}\, B$.

Thus, every local measurement at a single vertex is trapped between two dual constraints: Z_w is blinded by the unresolved orientation kernel ($\ker B$), while X_w is blinded by the settled edge coboundary ($\mathrm{im}\, B$). Neither can extract a bit locally.

4 Light Speed as the Rate of Homological Cycle Closure

If the relational parity y_{AB} is already settled in the spacelike spanning tree, what role does the relativistic speed of light c play in operational quantum mechanics?

Theorem 2.1 gives the exact answer: **on a tree ($m = n - 1$), the cycle space $\ker B^T$ has dimension $m - n + c = 0$.** A relationship on a tree can constrain future outcomes, but it can never be *verified* or *decoded* until a redundant edge closes a cycle ($\dim \ker B^T \geq 1$). In relativistic spacetime, creating that cycle-closing edge requires transporting classical or quantum records from the spacelike endpoints A and B along causal paths (at speed $\leq c$) to a common intersection event $M \in J^+(A) \cap J^+(B)$.

4.1 Theorem: The Relativistic Cycle-Closure Bound

Let $(M_{\text{spacetime}}, \preceq)$ be a causal poset (or Minkowski spacetime with causal order $p \preceq q \iff q \in J^+(p)$). A physical protocol consists of preparation events, local operations, and directed communication edges $e = (p \to q)$ which exist only if $p \preceq q$ (i.e., propagation speed $v \leq c$).

Theorem 4.1 (Relativistic Cycle-Closure Theorem)**.** *Suppose a source event S prepares an entangled pair with relational parity $y_{AB} \in \mathbb{F}_2$ and sends the subsystems along causal worldlines $e_1 = (S \to A)$ and $e_2 = (S \to B)$ to spacelike-separated measurement events A and B ($A \not\preceq B$ and $B \not\preceq A$). Let $x_A \in \mathbb{F}_2$ and $x_B \in \mathbb{F}_2$ be the local measurement outcomes recorded at A and B. Then:*

1. ***Zero Cycle Space Outside the Overlap:*** *At any spacetime event $P \notin J^+(A) \cap J^+(B)$, the causal subgraph accessible in the past light cone $J^-(P)$ contains at most one of $\{x_A, x_B\}$. The effective incidence graph on the observed endpoints has cycle-space dimension* $\dim \ker B^T = 0$*, and the conditional distribution of the unobserved parity check $x_A \oplus x_B \oplus y_{AB}$ is unresolvable at P.*

2. ***Minimum Cycle-Closure Time:*** *The earliest spacetime event M at which the homological cycle check*

$$\gamma_{SAB}: \quad x_A \oplus x_B \oplus y_{AB} = 0 \pmod 2 \tag{16}$$

can be evaluated lies on the boundary of the overlapping future light cone $\partial\big(J^+(A) \cap J^+(B)\big)$. If $A = (t_0, -L/2)$ and $B = (t_0, +L/2)$ are simultaneous in a Lorentz frame at spatial separation L, the cycle-space dimension transitions from $\dim \ker B^T = 0$ *to* $\dim \ker B^T = 1$ *at the exact light-crossing delay*

$$\Delta t_{\text{cycle}} = t_M - t_0 \geq \frac{L}{2c} \quad \textit{(at midpoint } M\textit{)}, \qquad \Delta t_{A \to B} \geq \frac{L}{c} \quad \textit{(at endpoint } B\textit{)}. \tag{17}$$

Proof. Verification of the relation (16) requires evaluating the linear functional $f(x_A, x_B, y_{AB}) = x_A \oplus x_B \oplus y_{AB}$. By Proposition 2.2 and Theorem 3.1, at any event $P \in J^+(A) \setminus J^+(B)$ (which has causal access to S and A, so y_{AB} and x_A are known, but no causal path from B), the variable x_B is separated from P. Even if P computes the predicted value $\hat{x}_B = x_A \oplus y_{AB}$, verifying that the actual record at B satisfies $x_B = \hat{x}_B$ (or detecting an edge error $e_{SB} = 1$ on branch $S \to B$) requires the third edge $e_3 = (B \to P)$ to close the 3-cycle $(S \to A \to P \leftarrow B \leftarrow S)$. Since causal edges only connect $q \to P$ when $q \in J^-(P)$, closing the cycle at P requires both $A \in J^-(P)$ and $B \in J^-(P)$, meaning $P \in J^+(A) \cap J^+(B)$. In $1 + 1$ Minkowski coordinates with $A = (t_0, -L/2)$ and $B = (t_0, +L/2)$, $J^+(A) \cap J^+(B) = \{(t, x) : c(t - t_0) \geq |x + L/2|$ and $c(t - t_0) \geq |x - L/2|\}$, whose earliest point is the midpoint $M = (t_0 + L/(2c), 0)$ and whose earliest intersection with Bob's worldline $x = +L/2$ is $t_B = t_0 + L/c$. □

4.2 Application: Quantum Teleportation as Pure Gauge-Bit Transport

Nothing illustrates Theorem 4.1 more clearly than standard quantum teleportation [6]. In popular accounts, teleportation is often described as "destroying a quantum state at A and reconstructing it at B using entanglement plus two classical bits."

Through the homological lens, what actually happens in quantum teleportation is far more surgical: **the continuous quantum amplitudes never travel across space from A to B at all; only the two unresolved discrete orientation bits ($\ker B$) travel at speed c to close the cycle and lift the $\rho_B = I/2$ gauge screen.**

Let us verify this explicitly:

1. **Initial Spanning Tree:** Alice holds a qubit C in an arbitrary unknown state $|\psi\rangle_C = \alpha|0\rangle + \beta|1\rangle$ (carrying two continuous real degrees of freedom on the Bloch sphere). Alice and Bob also share a Bell pair $|\Phi^+\rangle_{AB}$ prepared at source S in the past, which has settled relational stabilizers $Z_A Z_B = +1$ ($y_Z = 0$) and $X_A X_B = +1$ ($y_X = 0$). At this stage, the connected component is $\{A, B\}$ with 1 unresolved Z-orientation bit and 1 unresolved X-orientation bit.

2. **Alice's Relational Join at A:** Alice does *not* measure the state of C directly. Instead, she performs a joint Bell parity measurement on (C, A), measuring the two relational edge operators

$$S_{CA}^{(Z)} = Z_C \otimes Z_A \implies (-1)^{s_1}, \qquad S_{CA}^{(X)} = X_C \otimes X_A \implies (-1)^{s_2}, \tag{18}$$

   where $(s_1, s_2) \in \mathbb{F}_2^2$ are her two binary syndrome outcomes, each occurring with uniform probability $1/4$.

3. **Path Propagation Along the Spanning Tree:** Because (C, A) and (A, B) now form a two-edge spanning tree $C \overset{(s_1,s_2)}{\longleftrightarrow} A \overset{(0,0)}{\longleftrightarrow} B$, fixing the (C, A) edge parities to (s_1, s_2) immediately fixes the relative parity between C and B via XOR addition along the path ($s_1 \oplus 0 = s_1$ and $s_2 \oplus 0 = s_2$). Algebraic decomposition of the three-qubit state confirms that conditional on (s_1, s_2), Bob's qubit B is in the state

$$|\psi^{(s_1,s_2)}\rangle_B = X_B^{s_1} Z_B^{s_2} |\psi\rangle_B. \tag{19}$$

   Look at (19): the continuous amplitudes (α, β) are already present at B, up to a discrete Pauli gauge frame $X_B^{s_1} Z_B^{s_2}$ indexed by the two unresolved bits $(s_1, s_2) \in \mathbb{F}_2^2$!

4. **Why Bob is Completely Screened Before the Light Cone Arrives:** Prior to receiving (s_1, s_2) over the classical channel $A \to B$, Bob does not know which of the four gauge frames $\{I, X, Z, XZ\}$ holds. Averaging (19) over the uniform unresolved orientation group $\mathbb{F}_2^2$ yields the exact Pauli twirl identity:

$$\rho_B = \frac{1}{4} \sum_{(s_1,s_2)\in\mathbb{F}_2^2} \left(X^{s_1} Z^{s_2}\right) |\psi\rangle\langle\psi| \left(Z^{s_2} X^{s_1}\right) = \frac{I_2}{2}. \tag{20}$$

   Every trace of α and β is wiped out by the 2-bit orientation symmetry!

5. **Closing the Triangle at Speed c:** When Alice transmits the two syndrome bits (s_1, s_2) along a causal channel at speed $\leq c$ to Bob, that classical message serves as the **cycle-closing edge** $C/A \to B$. Upon arrival in $J^+(A)$, Bob applies $Z_B^{s_2} X_B^{s_1}$, cancelling the gauge orientation and recovering $|\psi\rangle_B$ with certainty.

TELEPORTATION IN ONE SENTENCE

Quantum teleportation does not squeeze continuous complex amplitudes (α, β) through a two-bit classical wire; the spanning tree $C - A - B$ fixes the relation immediately, while the two unresolved orientation bits $(s_1, s_2) \in \ker B$ screen Bob's local density matrix to $I_2/2$ until they arrive at the speed of light c to close the cycle.

5 Why Quantum Edges Outrank Classical Vertices: Obstructed Cycles

We now complete the argument by showing why the homological framework does not reduce quantum mechanics to a local hidden-variable theory.

In a purely classical graph over $\mathbb{F}_2$ (Section 2), the statement "the edge parity $y_{AB} = x_A \oplus x_B$ is settled while the global orientation bit $\ker B$ is unresolved" can always be interpreted epistemically: one could imagine that nature secretly picked one of the two vertex assignments ('01' or '10') at the source S, and we simply do not know which one.

Why does that epistemic interpretation fail in quantum mechanics? Because in the quantum stabilizer formalism (Section 3), **a single state can simultaneously satisfy cycle-closure equations in non-commuting bases whose joint vertex incidence matrix has no global solution in $\mathbb{F}_2$.**

Theorem 5.1 (Homological Obstruction of Simultaneous Vertex Assignments)**.** *Consider a 3-vertex cycle $G = K_3$ on qubits $\{A, B, C\}$ prepared in the 3-qubit GHZ stabilizer state $|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$, which is stabilized by the four commuting parity operators*

$$M_1 = X_A Y_B Y_C = -1, \quad M_2 = Y_A X_B Y_C = -1, \quad M_3 = Y_A Y_B X_C = -1, \quad M_4 = X_A X_B X_C = +1. \tag{21}$$

Write each local Pauli measurement outcome $(-1)^{x_v} \in \{+1, -1\}$ in terms of six hypothetical local vertex bits $z = (x_A, x_B, x_C, y_A, y_B, y_C)^T \in \mathbb{F}_2^6$. Then the four relational parity equations (21) form the linear system $M_{GHZ} z = w \pmod 2$:

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}}_{M_{GHZ} \in \mathbb{F}_2^{4\times 6}} \begin{pmatrix} x_A \\ x_B \\ x_C \\ y_A \\ y_B \\ y_C \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}}_{w \in \mathbb{F}_2^4} \pmod 2. \tag{22}$$

*The vector $\mathbf{1}_4 = (1,1,1,1)^T$ lies in the left cycle kernel $\ker M_{GHZ}^T$ ($\mathbf{1}_4^T M_{GHZ} = 0 \pmod 2$), whereas the right-hand syndrome satisfies $\mathbf{1}_4^T w = 1 \oplus 1 \oplus 1 \oplus 0 = 1 \neq 0 \pmod 2$. Consequently, the solution set $\{z \in \mathbb{F}_2^6 : M_{GHZ} z = w\}$ is **empty**: no pre-assigned local vertex values exist, even though every individual relational cycle equation in (21) is satisfied with probability 1 when measured and verified in the future light cone $J^+(A) \cap J^+(B) \cap J^+(C)$.*

Proof. In the stabilizer group of $|\mathrm{GHZ}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$, we have $X_A X_B X_C |\mathrm{GHZ}\rangle = +|\mathrm{GHZ}\rangle$ and $Z_u Z_v |\mathrm{GHZ}\rangle = +|\mathrm{GHZ}\rangle$ for all pairs. Since $Y_B Y_C = (iX_B Z_B)(iX_C Z_C) = -X_B X_C Z_B Z_C$, we obtain $X_A Y_B Y_C = -(X_A X_B X_C)(Z_B Z_C)$, which acts on $|\mathrm{GHZ}\rangle$ with eigenvalue $-(+1)(+1) = -1$ (parity bit 1 in $\mathbb{F}_2$). By symmetry, M_1, M_2, M_3 have parity bit 1 and M_4 has parity bit 0, giving $w = (1,1,1,0)^T$. Summing the four rows of M_{GHZ} over $\mathbb{F}_2$ counts each variable $x_A, x_B, x_C, y_A, y_B, y_C$ exactly twice, so $\mathbf{1}_4^T M_{\mathrm{GHZ}} = (2,2,2,2,2,2) \equiv (0,0,0,0,0,0) \pmod 2$. Thus any $w \in \mathrm{im}\, M_{\mathrm{GHZ}}$ must satisfy the cycle consistency condition $\mathbf{1}_4^T w = 0 \pmod 2$ (Theorem 2.1, Part 2). Since $\mathbf{1}_4^T w = 1 \pmod 2$, $w \notin \mathrm{im}\, M_{\mathrm{GHZ}}$. □

Theorem 5.1 is Mermin's GHZ parity proof [5] expressed in the exact language of Section 05 of [9]:

- In Section 05 of [9], a classical triangle with $\bigoplus_{e\in\gamma} y_e = 1$ indicated a **corrupted edge report** relative to a pre-existing vertex assignment $x \in \mathbb{F}_2^n$.
- In Theorem 5.1, the quantum state produces verified relational parities with $\mathbf{1}_4^T w = 1$ in the absence of any error, proving that **the relational stabilizers are primary and no underlying vertex assignment $z \in \mathbb{F}_2^6$ exists prior to measurement**.

Whenever Alice, Bob, and Charlie choose to measure one of the four rows of (22)—say Row 1 (X_A, Y_B, Y_C)—their local operators commute on that row, restricting the system to a classical 3-vertex hyperedge where rank $= 1$, leaving 2 unresolved local orientation bits ($\rho_A = \rho_B = \rho_C = I_2/2$) that screen all faster-than-light signaling until their three classical bits meet in $J^+(A) \cap J^+(B) \cap J^+(C)$ to verify $x_A \oplus y_B \oplus y_C = 1$.

6 Synthesis Ledger and Conclusion

Table 2 unifies the mathematical structure across classical symbolic dynamics, graph parity networks, relativistic causal diamonds, and quantum stabilizer codes.

Domain	**System / Operator**	**Resolved Structure (im B)**	**Unresolved Space (**ker B**)**	**Cycle Closure (**ker B^T**)**
01 / Graph Parity	Incidence $B \in \mathbb{F}_2^{m\times n}$	Spanning forest parities ($n-c$ bits)	Component roots / global flip (c bits)	Fundamental cycles ($m-n+c$ checks)
02 / Spacelike EPR	Open Tree ($n=2, m=1$)	Past edge parity $y_{AB} = x_A \oplus x_B$ (1 bit)	Global bit flip $\{01, 10\}$ (1 bit; $\rho_B = I/2$)	$\dim\ker B^T = 0$ (No local check possible)
03 / Light-Cone Merge	Triangle in $J^+(A) \cap J^+(B)$	Path parities y_{SA}, y_{SB}, y_{AB} (2 bits)	Global origin root $x_S \in \mathbb{F}_2$ (1 bit)	$\dim\ker B^T = 1$: $x_A \oplus x_B \oplus y_{AB} = 0$
04 / Teleportation	2-Edge Tree $C - A - B$	Bell syndromes (s_1, s_2) fixing $X^{s_1}Z^{s_2}\lvert\psi\rangle_B$	Unresolved Pauli frame (2 bits; twirls to $I/2$)	Classical wire $A \to B$ at speed $\le c$
05 / GHZ Paradox	Symplectic $M_{\text{GHZ}} \in \mathbb{F}_2^{4\times 6}$	4 commuting 3-body stabilizer parities	2 bits per measured row ($\rho_v = I/2$)	Global obstruction $\mathbf{1}_4^T w = 1$ (no local z)

Table 2: Unified homological ledger across classical graphs, relativistic light cones, and quantum stabilizer states.

6.1 Closing Reflection: Why the Unresolved Bit Protects Causality

When we ask how two entangled particles can "know" to agree or disagree across ten light-years without sending a faster-than-light signal, the question presupposes that the universe must store its state in the vertices rather than the edges. Once we drop that assumption, the mystery evaporates:

1. **Relationships are stored in the cut space (im B):** They are established locally when subsystems interact in their shared causal past. No relationship ever needs to jump across a spacelike interval.

2. **Causality is protected by the orientation kernel (**ker B**):** Because fixing every relationship on a connected component always leaves one global orientation bit unresolved, a local observer holding one endpoint of an entangled state holds half of a parity equation—which is locally indistinguishable from a fair coin ($\rho_B = I/2$).

3. **Light speed governs the cycle space (**ker B^T**):** Converting a settled relationship into a verified fact or an operational signal requires bringing both endpoints of the parity equation together to close a homological triangle. That third edge must travel through spacetime at or below the speed of light c.

Seen in this light, the unresolved bit is not a limitation of our measurement apparatus or an incomplete chapter in quantum theory. It is the exact structural symmetry that allows a universe to maintain global relational coherence while respecting local relativistic causality.

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